

- Q.28 Enlist the various properties of Aluminium? Also name various alloys of Aluminium ?
- Q.29 Write at least five properties of High carbon steel.
- Q.30 Differentiate between slip and Twinning.
- Q.31 Explain
- Nuclear Materials
 - Spring Materials
- Q.32 Write two effects of alloying Carbon and Phosphorus.
- Q.33 Define thermoplastics with its types.
- Q.34 Explain Heat Insulating materials.
- Q.35 Differentiate between metal and non-metals.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain various mechanical properties of metals.
- Q.37 Explain the classification and properties of Ceramics and composites in brief.
- Q.38 Explain the classification of plain carbon steels in details.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The element which can replace one or more atoms of Hydrogen from an acid is called
- Metal
 - Metalloid
 - Non-metal
 - Alloy
- Q.2 The ability of material to reflect light when finely polished is called
- Colour
 - Density
 - Luster
 - Hardness
- Q.3 Cast Iron has property of
- Ductility
 - Brittleness
 - Plasticity
 - Elasticity
- Q.4 In B.C.C structure, the number of atoms in unit cell are
- 2
 - 1
 - 6
 - 4

Q.5 Which of the following is an amorphous material?

- a) Glass b) Mica
- c) Brass d) Copper

Q.6 Which of the following is an alloy?

- a) Gold b) Silver
- c) Aluminium d) Brass

Q.7 Bronzes are the alloy of

- a) Copper-tin b) Copper-nickel
- c) Copper-Nickel-zinc d) Copper-Zinc

Q.8 In annealing process of heat treatment, cooling of steel is carried in

- a) Air b) Oil
- c) Furnace d) None of the above

Q.9 The full form of P.V.C is

- a) Polyvinyl formaldehyde
- b) Polyvinyl Carbonate
- c) Prevynyl chloride
- d) Polyvinyl chloride

Q.10 The trade name of polystyrene is

- a) Thermocole b) Mica
- c) Cork d) Glass wool

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define non-ferrous metals.

Q.12 Name any two Semi-Conductors.

Q.13 Define Unit cell.

Q.14 Define Thermo setting plastics.

Q.15 Name different types of Ceramics.

Q.16 Name the purest form of Iron.

Q.17 Define normalizing.

Q.18 Define Composite material?

Q.19 Give the industrial application of silicon steel.

Q.20 Define slip.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain types of Heat treatment in brief?

Q.22 Write the purposes of Heat treatment.

Q.23 Define Iron and its types.

Q.24 Write various uses of High Speed steel.

Q.25 Give the classification of Engineering materials?

Q.26 Write the advantages of Semi-Conductors.

Q.27 Write the properties of Glass wool.